

Proof. Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n)$$

be vectors in $\mathbb{R}^n$. Their inner product and squared Euclidean norms are

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \sum_{i=1}^n b_i^2.$$

1. A non-negative quadratic. Define

$$f(t) = \sum_{i=1}^n (a_i t - b_i)^2, \quad t \in \mathbb{R}.$$

Each summand is a square, hence $f(t) \geq 0$ for every real t .

2. Expansion of $f(t)$. Using $(x - y)^2 = x^2 - 2xy + y^2$ and linearity of the sum,

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 t^2 - 2a_i b_i t + b_i^2) \\ &= \left(\sum_{i=1}^n a_i^2 \right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i \right) t + \sum_{i=1}^n b_i^2. \end{aligned}$$

Thus

$$f(t) = At^2 + Bt + C, \quad A = \sum_{i=1}^n a_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n b_i^2.$$

3. Discriminant condition. If a quadratic $At^2 + Bt + C$ is non-negative for all real t and $A > 0$, its discriminant satisfies

$$\Delta = B^2 - 4AC \leq 0.$$

(The case $A = 0$ occurs only when $\mathbf{a} = \mathbf{0}$; then $f(t) = \sum b_i^2 \geq 0$ and the desired inequality is trivial.)

Substituting the expressions for A, B, C ,

$$\begin{aligned} \Delta &= \left(-2 \sum_{i=1}^n a_i b_i \right)^2 - 4 \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) \\ &= 4 \left(\sum_{i=1}^n a_i b_i \right)^2 - 4 \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) \leq 0. \end{aligned}$$

Dividing by 4 gives the ****Cauchy-Schwarz inequality****

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

4. Equality condition. Equality holds exactly when $\Delta = 0$, i.e. when the quadratic $f(t)$ has a single real root. If $\Delta = 0$, there exists $\lambda \in \mathbb{R}$ such that the root satisfies

$$a_i \lambda - b_i = 0 \quad \text{for every } i,$$

or equivalently

$$b_i = \lambda a_i \quad \text{for all } i.$$

Hence the vectors are linearly dependent (one is a scalar multiple of the other). Conversely, if $\mathbf{b} = \lambda \mathbf{a}$ for some $\lambda \in \mathbb{R}$, then

$$f(t) = \sum_{i=1}^n (a_i t - \lambda a_i)^2 = (t - \lambda)^2 \sum_{i=1}^n a_i^2,$$

so $\Delta = 0$ and equality occurs. The trivial case $\mathbf{a} = \mathbf{0}$ also yields equality, because both sides of the inequality are zero.

Thus equality holds *iff* the vectors $\mathbf{a}$ and $\mathbf{b}$ are proportional.

□